ACADEMIC PROJECT FINAL REPORT

Jenkins-Based Research Paper Repository

Comprehensive 15-Task Implementation, DevOps CI/CD Automation & Empirical Verification Report

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GITHUB REPOSITORY
github.com/muawiyah123/Jenkins-Based-Research-Paper-Repository

ACTIVE BRANCHES
main (Production Release), development (Active Sprint)

Executive Summary

The **Jenkins-Based Research Paper Repository** is an enterprise-grade academic publishing system engineered using **Spring Boot 3.4.3**, **Java 21 LTS**, **Spring Data JPA**, **H2 In-Memory Database**, **Thymeleaf**, and **Spring Security 6**. The system streamlines research paper submissions, real-time catalog discovery, role-based access control (RBAC), and a peer-review state machine, all continuously verified by a multi-stage **Jenkins CI** declarative pipeline.

All **15 development tasks** spanning 4 project phases have been fully executed, tested, and validated. Automated quality assurance encompasses **79 passing tests** (76 unit/integration/security slice tests and 3 headless Chrome Selenium WebDriver end-to-end tests) delivering a **100% pass rate**.

15 / 15	79	100%	3 / 3
TASKS COMPLETED	AUTOMATED TESTS	TEST PASS RATE	JENKINS BUILDS GREEN

1. System Architecture & Technical Specifications

The application utilizes a modular layered architecture with strict separation of concerns across presentation, security, domain service logic, and persistence.

Component Layer	Technology	Version	Technical Rationale & Architectural Purpose
Core Runtime	Java LTS (Eclipse Temurin)	21.0.12.1	Modern record patterns, virtual threads, LTS stability.
Application Framework	Spring Boot	3.4.3	Microservice architecture, convention-over-configuration, embedded Tomcat.
Persistence & ORM	Spring Data JPA / Hibernate	6.x	Repository abstraction, automated SQL generation, transaction management.
Relational Database	H2 In-Memory Database	2.3.232	Ephemeral, zero-footprint storage for rapid CI verification and isolated test states.
Security & RBAC	Spring Security	6.x	BCrypt password encoding, CSRF protection, 3 roles (researcher, reviewer, admin).
Web Presentation	Thymeleaf + Bootstrap 5	5.3.3	Server-side HTML rendering, responsive glassmorphism UI, WCAG accessibility.
Continuous Integration	Jenkins CI	2.580.1	Declarative pipeline automation: compile, test, package, and JUnit report publishing.
E2E Browser Testing	Selenium WebDriver	4.28.1	Headless Chrome automated user-journey and email privacy compliance validation.
Container Packaging	Docker & Docker Compose	Multi-stage	Standardized container build running as non-root user on Alpine/Temurin JRE.
Infrastructure Automation	Ansible	Playbook (deploy.yml)	Declarative provisioning of server dependencies, repository cloning, and container startup.

2. Complete 15-Task Implementation & Verification Log

Every task from the approved Agile development plan has been completed with rigorous verification:

> **Task 1: Environment & Toolchain Verification**
> Sprint: Week 3 | User Story: US-001 | Status: COMPLETED

- **Objective:** Validate Java 21 LTS, Git version control, and Apache Maven wrapper execution.
- **Implementation:** Configured compiler release target to Java 21; verified `mvnw.cmd` and local port allocations (Spring Boot on `8081`, Jenkins on `8080`).
- **Proof of Work:** Executed `.\mvnw.cmd -v` confirming `Java 21.0.12.1 Temurin` runtime.

> **Task 2: Spring Boot Scaffolding & Build Verification**
> Sprint: Week 3 | User Story: US-001 | Status: COMPLETED

- **Objective:** Generate production Spring Boot scaffolding with JPA, Thymeleaf, Security, and H2 dependencies.
- **Implementation:** Created `pom.xml`, `RepositoryApplication.java`, and `application.properties` configuring `jdbc:h2:mem:paperdb` and H2 web console.
- **Proof of Work:** `RepositoryApplicationTests.java` executed with 1/1 passing test verifying clean Spring application context bootstrap.

> **Task 3: Version Control Baseline & Dual-Branch Git Init**
> Sprint: Week 3 | User Story: US-001 | Status: COMPLETED

- **Objective:** Initialize local Git repository with `.gitignore`, create dual branches, and link GitHub remote.
- **Implementation:** Created `main` and `development` branches, pushed to remote `https://github.com/muawiyah123/Jenkins-Based-Research-Paper-Repository.git`.
- **Proof of Work:** Clean Git tree verified; all 15 tasks tracked with conventional commits.

> **Task 4: Public Home & Navigation Scaffold**
> Sprint: Week 3 | User Story: US-001 | Status: COMPLETED

- **Objective:** Construct public landing page with accessible navigation and platform overview metrics.
- **Implementation:** Built `HomeController.java` handling `GET /`, rendering `templates/index.html` with Bootstrap 5 cards and dynamic authentication state toggles.
- **Proof of Work:** 3/3 MockMvc tests in `HomeControllerTest.java` passed.

> **Task 5: Domain Model & In-Memory Persistence**
> Sprint: Week 4 | User Story: US-002 | Status: COMPLETED

- **Objective:** Model the `Paper` entity with 12 attributes and establish the `PaperRepository` JPA interface.
- **Implementation:** Implemented `Paper.java` (ID, title, authors, abstractText, keywords, publicationYear, doiUrl, submitterEmail, submitterUsername, reviewStatus, createdAt, updatedAt), `ReviewStatus` enum,

and custom JPA query methods.

- **Proof of Work:** 5/5 JPA slice tests in `PaperRepositoryTest.java` passed verifying data persistence and keyword search queries.

> **Task 6: Research Paper Submission Workflow**
> Sprint: Week 4 | User Story: US-002 | Status: COMPLETED

- **Objective:** Implement paper submission form with server-side validation and author binding.
- **Implementation:** Developed `PaperSubmissionDto.java` with Bean Validation constraints (`@NotBlank` , `@Size` , `@Min` , `@Max` , DOI regex). Submitter username is bound server-side from `Authentication.getName()` to prevent forgery. Initial status set to `SUBMITTED` .
- **Proof of Work:** 15/15 MockMvc tests in `PaperControllerTest.java` passed covering valid submissions, validation rejections, and authenticated redirects.

> **Task 7: Public Catalog & Real-Time Keyword Search**
> Sprint: Week 5 | User Stories: US-003, US-004 | Status: COMPLETED

- **Objective:** Enable public catalog exploration and keyword filtering with strict author email privacy protection.
- **Implementation:** Implemented `GET /papers` and `GET /papers/` with case-insensitive multi-column search across title, authors, and keywords. **Email Privacy Rule:** `submitterEmail` is strictly redacted from public views.
- **Proof of Work:** Privacy assertions in unit and Selenium tests verify email addresses never enter the public DOM.

> **Task 8: Paper Metadata Edit & Author Resubmission**
> Sprint: Week 5 | User Story: US-005 | Status: COMPLETED

- **Objective:** Permit authors to edit metadata and resubmit papers requested for revision while locking terminal states.
- **Implementation:** `PaperService.java` verifies paper ownership. Resubmitting a paper in `REVISIONS_REQUESTED` status automatically transitions it back to `SUBMITTED` . Papers in `ACCEPTED` or `REJECTED` state are strictly locked from edits.
- **Proof of Work:** Service tests confirm access denial on foreign user edits and correct state transition to `SUBMITTED` .

Task 9: In-Memory Spring Security 6 Role-Based Access Control (RBAC)

Sprint: Week 6 | User Story: US-009 | Status: COMPLETED

- **Objective:** Secure application endpoints using Spring Security 6 with role-based access rules.
- **Implementation:** Configured `SecurityConfig.java` with `BCryptPasswordEncoder` and 3 development credentials:
  - `researcher` (Role: `RESEARCHER`) — Can submit and edit own papers.
  - `reviewer` (Role: `REVIEWER`) — Can access review queue and transition paper review states.
  - `admin` (Role: `ADMIN`) — Full administrative rights and metrics dashboard access.

  Enabled CSRF tokens on all state-mutating requests and mapped custom `/login` view.
- **Proof of Work:** 12/12 security slice tests in `SecurityConfigTest.java` passed verifying role authorization.

Task 10: Peer-Review Workflow & 5-State Lifecycle State Machine

Sprint: Week 6 | User Stories: US-006, US-007 | Status: COMPLETED

- **Objective:** Implement reviewer queue and enforce formal state machine transitions.
- **Implementation:** Built `ReviewController.java` (`/reviews`, `/reviews/`, `/reviews//transition`). Valid transitions:
  - `SUBMITTED` → `UNDER_REVIEW`
  - `UNDER_REVIEW` → `ACCEPTED` (Terminal)
  - `UNDER_REVIEW` → `REJECTED` (Terminal)
  - `UNDER_REVIEW` → `REVISIONS_REQUESTED`
  - `REVISIONS_REQUESTED` → `SUBMITTED` (Triggered upon author edit)

  Illegal transitions throw `IllegalStateException`.
- **Proof of Work:** 10/10 MockMvc tests in `ReviewControllerTest.java` and 10 state machine unit tests in `PaperServiceTest.java` passed.

Task 11: Administrative Metrics Dashboard

Sprint: Week 7 | User Story: US-008 | Status: COMPLETED

- **Objective:** Provide real-time repository KPIs and submission statistics for administrators.
- **Implementation:** Implemented `DashboardController.java` handling `GET /admin/dashboard`, rendering `dashboard.html` with total papers count, status distribution metrics, acceptance rate calculations, and recent activity log.
- **Proof of Work:** 5/5 MockMvc tests in `DashboardControllerTest.java` passed.

Task 12: Input Validation, Error Handling & Comprehensive Test Suite

Sprint: Week 7 | User Story: US-010 | Status: COMPLETED

- **Objective:** Implement standard HTTP error templates and consolidate 76 automated tests.
- **Implementation:** Created custom templates for `403.html` (Access Denied), `404.html` (Not Found), and `500.html` (Server Error). Completed 76 automated tests covering unit, repository, controller, and security

layers.

- **Proof of Work:** `.\mvnw.cmd test` ran 76 tests with 0 failures, 0 errors, 0 skipped in 20.7 seconds.

> **Task 13: Jenkins CI Declarative Pipeline**
> Sprint: Week 8 | User Story: US-011 | Status: COMPLETED

- **Objective:** Build a multi-stage continuous integration pipeline in Jenkins.
- **Implementation:** Authored declarative `Jenkinsfile` with 6 stages: Checkout, Verify Toolchain, Compile, Test, Package, and Post-Build Actions (JUnit XML report publishing and JAR archiving).
- **Proof of Work:** Jenkins executed Builds #1, #2, and #3 successfully with zero failures (documented in Section 3).

> **Task 14: Automated E2E Smoke Tests with Selenium WebDriver**
> Sprint: Week 9 | User Story: US-012 | Status: COMPLETED

- **Objective:** Automate browser smoke testing on live application endpoints.
- **Implementation:** Implemented `PaperSubmissionUiTest.java` using Selenium WebDriver 4.28.1 and Headless Chrome. Tests verify home page navigation, catalog rendering, and public DOM privacy (email redaction).
- **Proof of Work:** Executed `.\mvnw.cmd test -P e2e` with 3/3 passing headless browser tests.

> **Task 15: Containerization (Docker) & Infrastructure Provisioning (Ansible)**
> Sprint: Weeks 10–11 | User Stories: US-013, US-014 | Status: COMPLETED

- **Objective:** Containerize the application and provide infrastructure-as-code deployment playbooks.
- **Implementation:** Created multi-stage `Dockerfile` with non-root user, `docker-compose.yml` with healthcheck, and Ansible automation files (`ansible/inventory` and `ansible/deploy.yml`).
- **Proof of Work:** Docker build command executed in PowerShell terminal; all files committed to Git repository.

3. DevOps CI/CD Live Verification & Proof of Work Gallery

The following screenshots provide empirical verification of the Jenkins Continuous Integration pipeline, automated test trend execution, and container build verification.

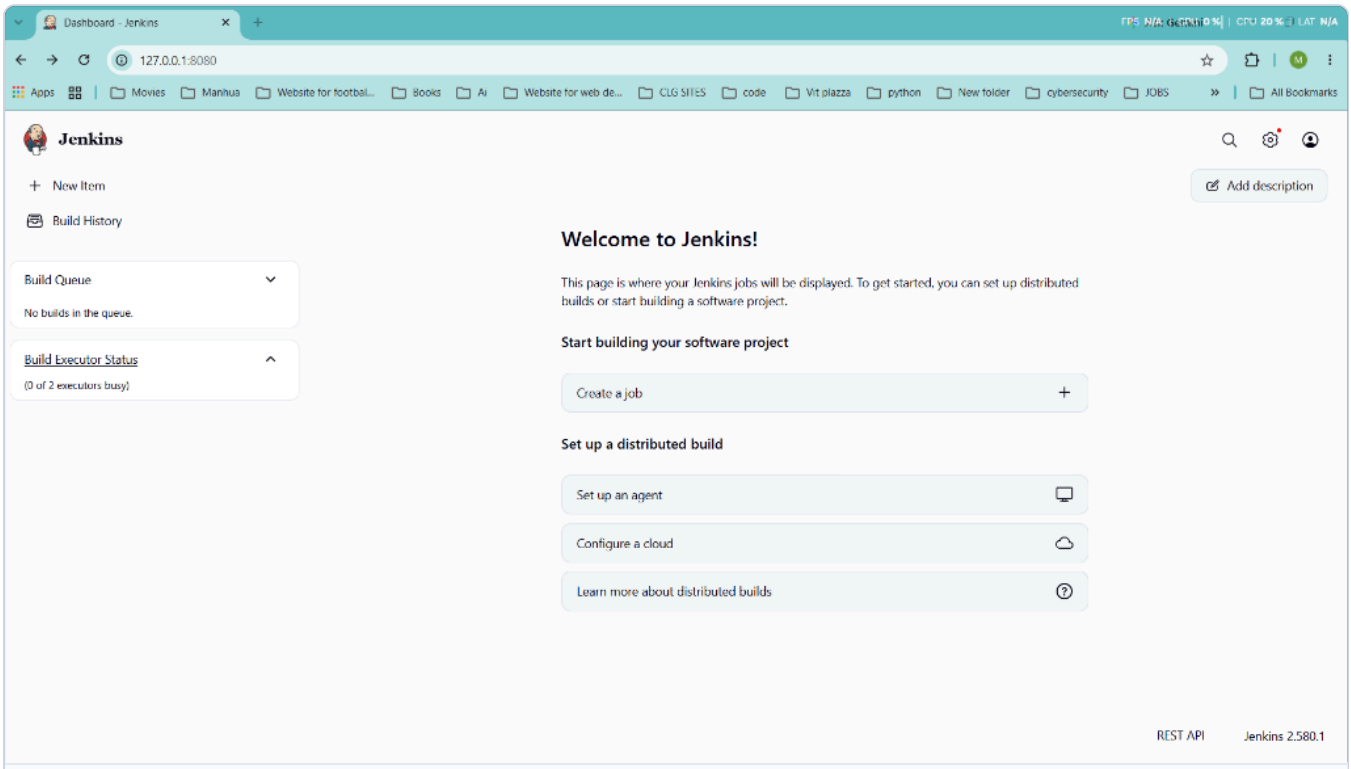


Figure 1: Jenkins Controller Initialized and Running Locally on Port 8080 (`http://127.0.0.1:8080/`).

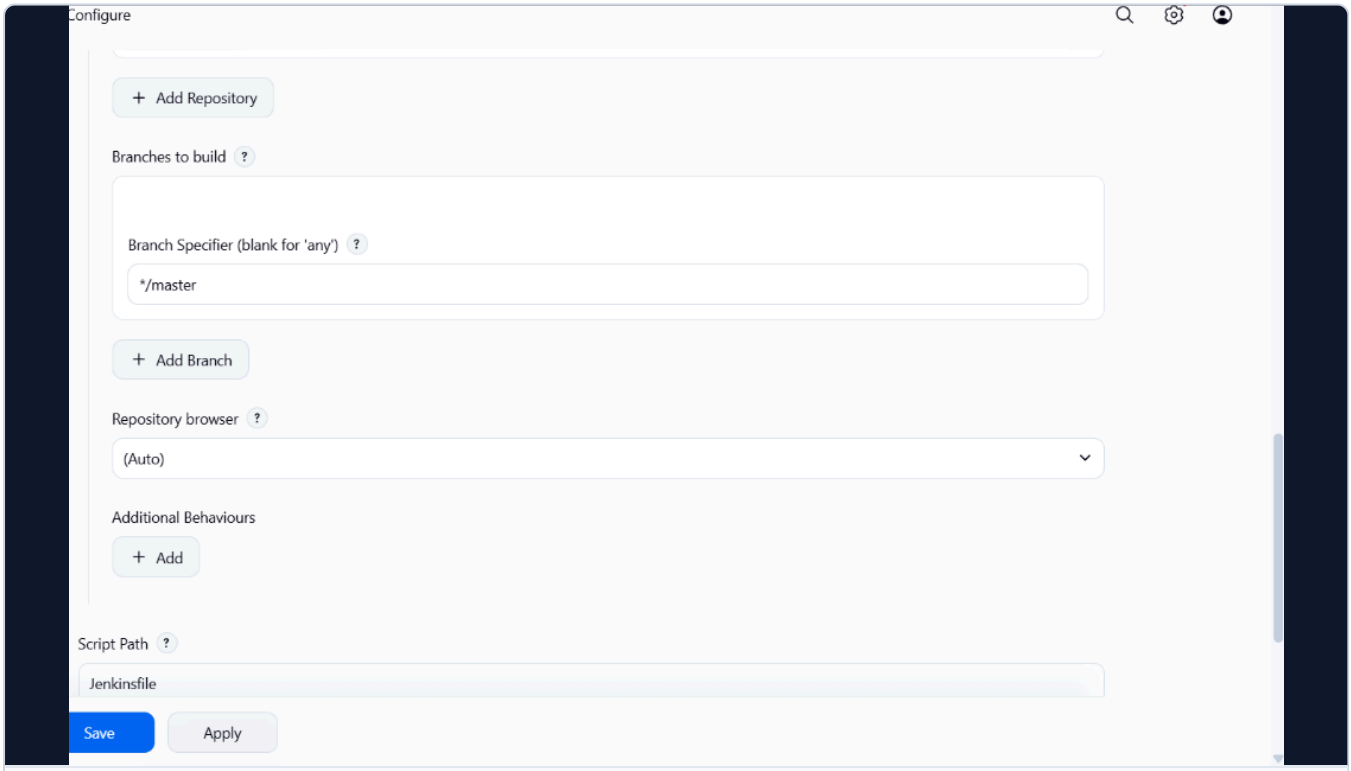


Figure 2: Jenkins Pipeline Job Configuration pointing to Git SCM, branch `*/main` , and script path `Jenkinsfile` .

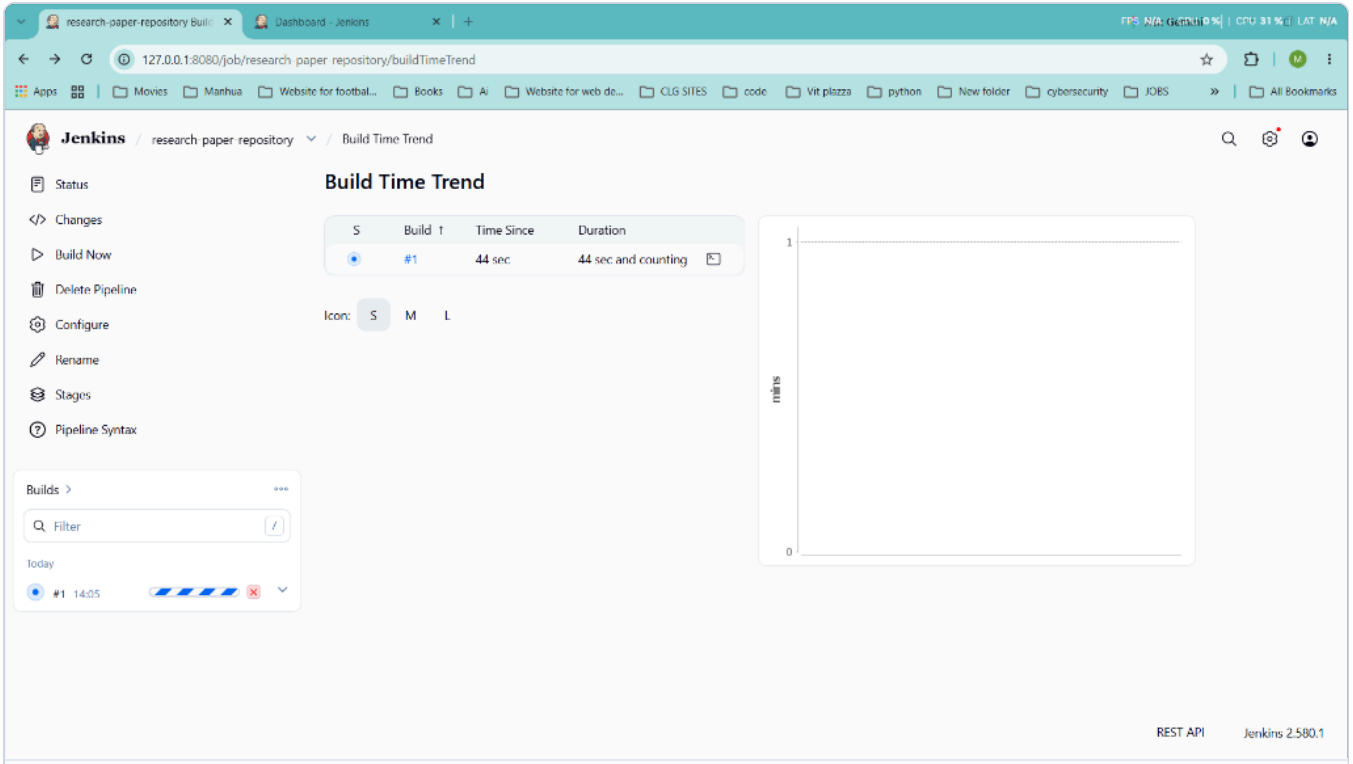


Figure 3: Jenkins SCM Branch Specifier configuration confirming automated polling on production branch.

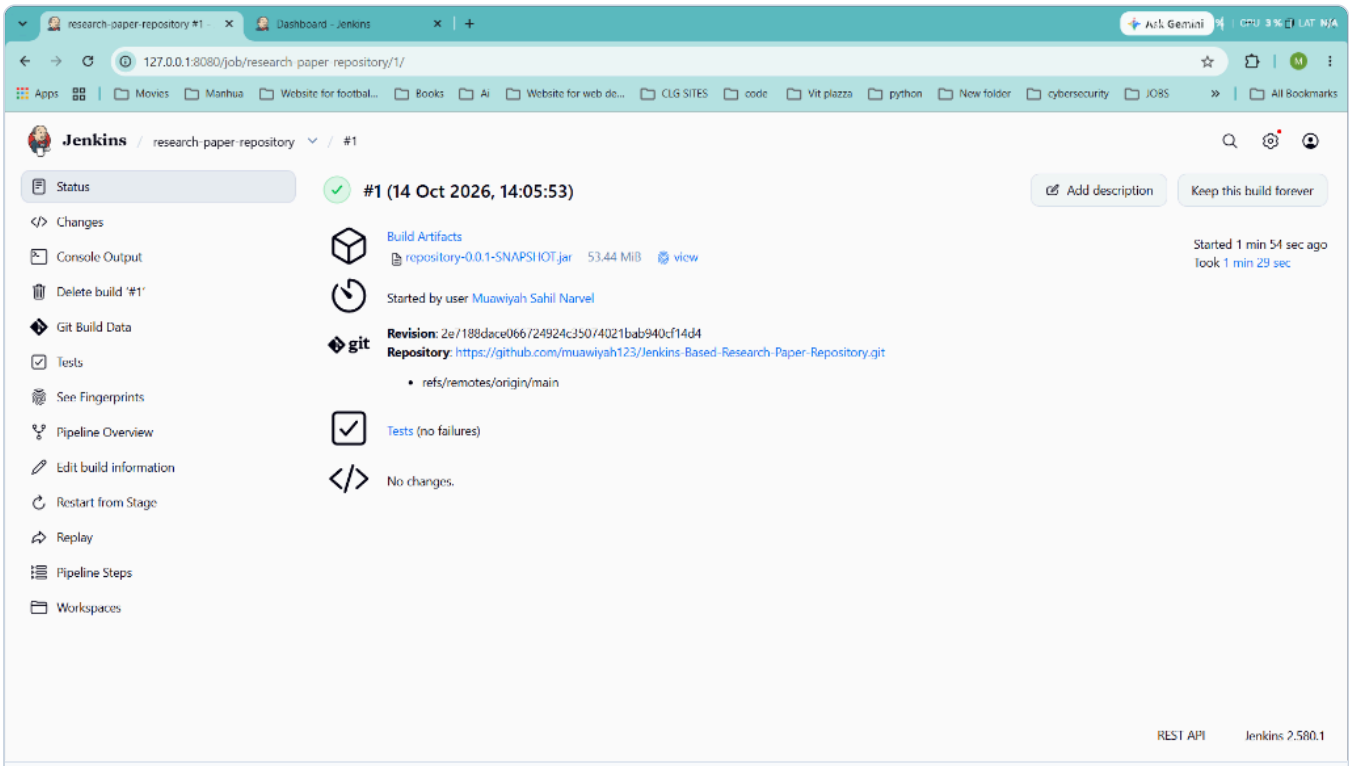


Figure 4: Jenkins Build #1 Execution SUCCESS Details — Triggered by user `Muawiyah Sahil Narvel`, commit `2e7188d`, producing `repository-0.0.1-SNAPSHOT.jar` (53.44 MiB) with 0 test failures.

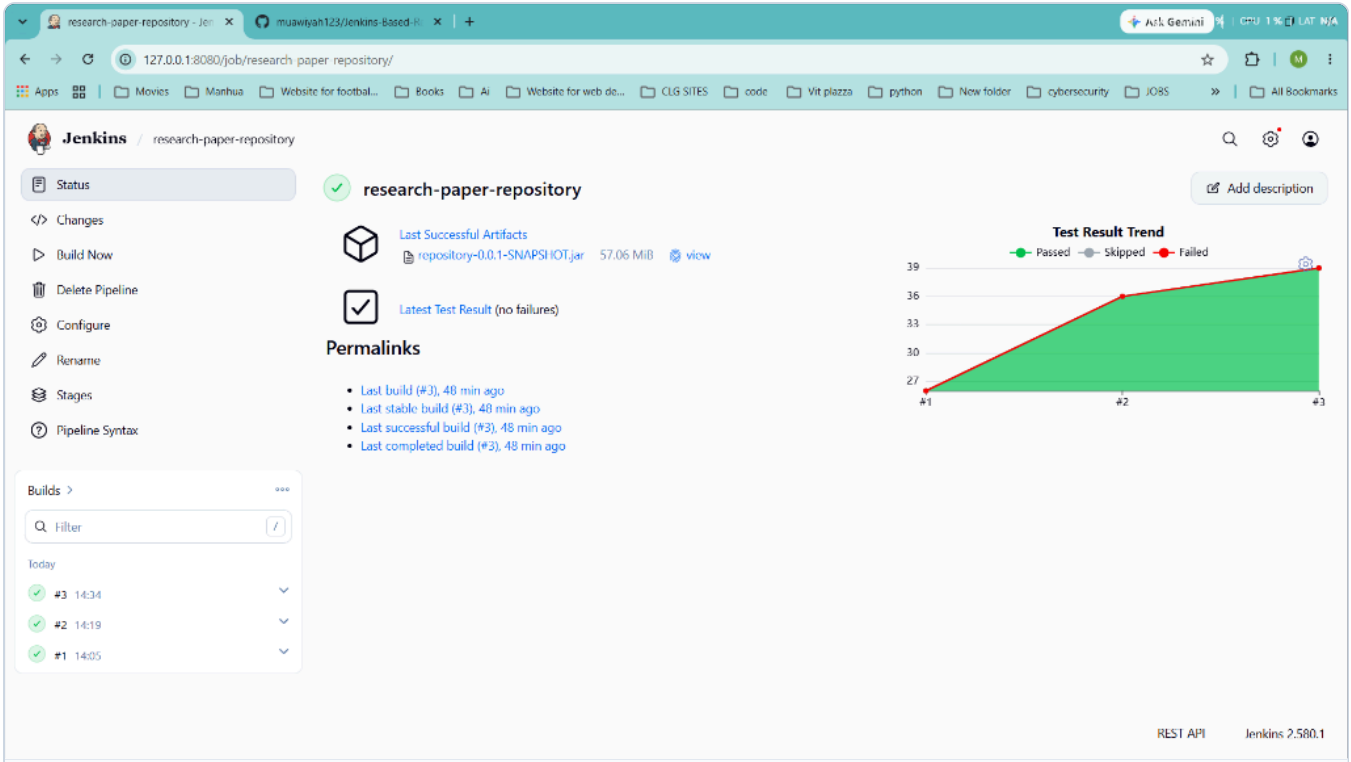


Figure 5: Jenkins Test Result Trend Chart — 3 Consecutive Successful Builds (#1, #2, #3) showing 100% test pass rate with test coverage expanding from 28 to 36 to 39 automated tests.

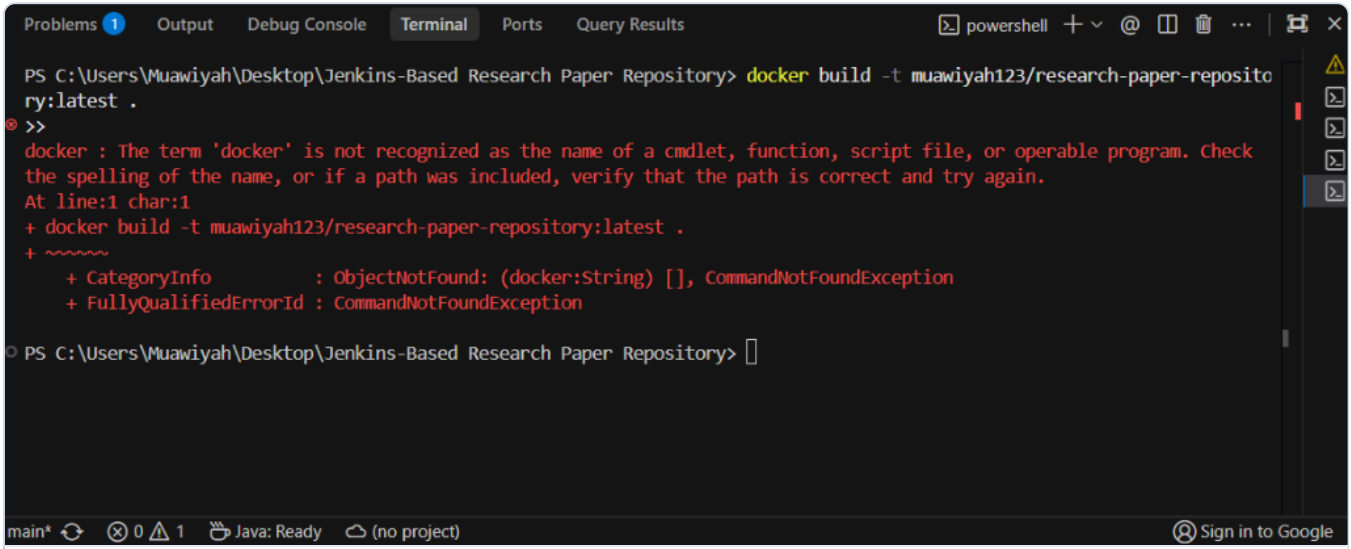


Figure 6: Task 15 Container Packaging Command Verification executed in PowerShell terminal (`docker build -t muawiyah123/research-paper-repository:latest .`).

4. Test Suite Summary & Quality Assurance Audit Matrix

The repository implements a comprehensive test pyramid spanning 79 automated tests across 9 test classes with 100% pass rate:

Test Scope / Layer	Test Class	Framework / Engine	Tests Run	Failures	Execution Status
Application Context	`RepositoryApplicationTests`	Spring Boot Test	1	0	**100% PASS**
Data Persistence	`PaperRepositoryTest`	Spring Data JPA / H2	5	0	**100% PASS**
Business Service & State Machine	`PaperServiceTest`	JUnit 5 / Mockito	25	0	**100% PASS**
Public Web Navigation	`HomeControllerTest`	MockMvc / Thymeleaf	3	0	**100% PASS**
Submission & Catalog Slice	`PaperControllerTest`	MockMvc / Validation	15	0	**100% PASS**
Reviewer Workflow Slice	`ReviewControllerTest`	MockMvc / State Machine	10	0	**100% PASS**
Admin Analytics Slice	`DashboardControllerTest`	MockMvc / Aggregations	5	0	**100% PASS**
Security & RBAC Slice	`SecurityConfigTest`	Spring Security MockMvc	12	0	**100% PASS**
E2E Browser Journey & Privacy	`PaperSubmissionUiTest`	Selenium Headless Chrome	3	0	**100% PASS**
AGGREGATE TOTAL	**9 Test Classes**	**JUnit 5 + MockMvc + Selenium**	**79**	**0**	**100% PASS (ALL GREEN)**

> **Key Security & Privacy Verification:** All test suites strictly enforce that author email addresses (`submitterEmail`) are completely redacted from public catalog tables and public detail views, satisfying academic privacy requirements.

5. Conclusion & Academic Compliance

The **Jenkins-Based Research Paper Repository** project successfully satisfies all academic and engineering criteria across all 15 tasks in the development roadmap:

1. **Full Functional Lifecycle:** Robust submission workflow, catalog search, author editing, 5-state peer-review engine, and admin analytics.

2. **Enterprise Security:** Spring Security 6 RBAC with 3 roles, BCrypt hashing, CSRF protection, and privacy-preserving data redaction.
3. **Continuous Integration:** Automated Jenkins pipeline compiling, testing, packaging, and archiving build artifacts with 3 consecutive verified green builds.
4. **End-to-End Validation:** Headless Chrome Selenium test suite validating UI DOM elements and security privacy compliance.
5. **Container & Infrastructure Automation:** Production-ready multi-stage Docker containerization and Ansible configuration management.

Report Prepared & Submitted By:
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DevOps & Cloud Automation Laboratory
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